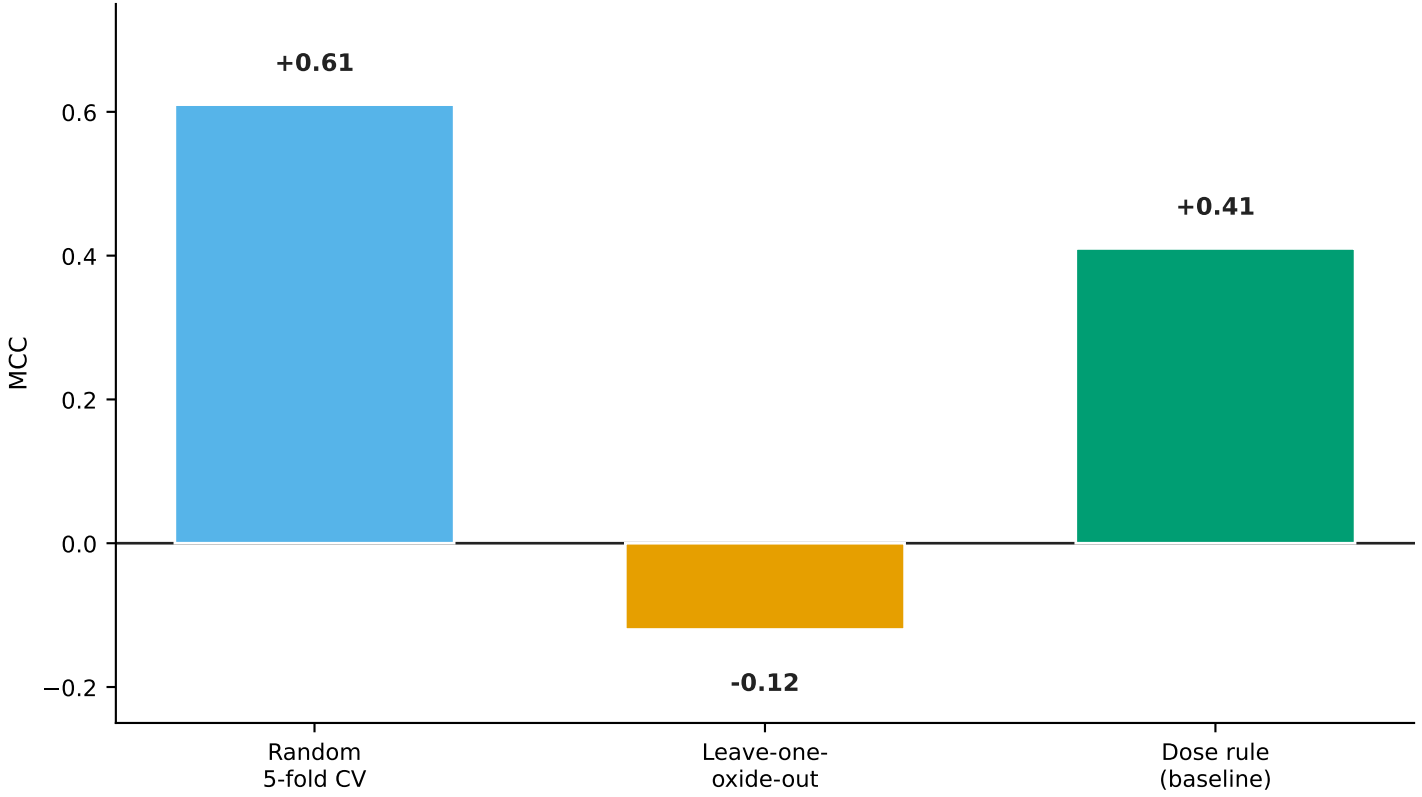
Validation strategy governs apparent ML performance

Random-split vs material-level evaluation, and a dose baseline



Key findings

Random splitting inflates performance; the gap grows with duplicate records.

Holding out whole oxides drops performance to chance.

A one-variable dose rule matches or beats all five ML models.